

Wazuh SIEM

Setup and Configuration

Project description

I was tasked with setting up a Security Information and Event Management (SIEM) tool to practice log analysis, threat detection. To complete this task, I performed the following steps to deploy the environment and ingest data:

Deploying and Configuring the Virtual Machine

I downloaded VMware workstation and imported a pre-built Wazuh .ova virtual machine file.

Accessing the Wazuh Dashboard

I started the VM and ran the `ifconfig` command to find the VM's IP address. Using this IP address, I navigated to the Wazuh web interface in my browser (`https://the_local_ip_address`) and logged into the dashboard using the default admin credentials.

Ingesting and Analyzing Data

To get the data into the SIEM for analysis, I went to the host's terminal and hosted a python http server where the tutorial data is located :

```
python -m http.server 4444
```

I connected to the VM through SSH.

```
ssh wazuh-user@local_vm_ip
```

Then I proceeded to wget the .zip file called tdata.zip

```
wget http://local_host_ip:4444/tdata.zip
```

```
mkdir tdata
```

```
mv tdata.zip tdata/
```

```
cd /tdata/
```

Finally, I unzipped the file.

```
unzip tdata.zip
```

I used the vim text editor to create a configuration file named ingest.yml.

```
seccomp.enabled: false
filebeat.inputs:
- type: log
  enabled: true
  paths:
    - /home/wazuh-user/tdata/**/*.*log
    - /home/wazuh-user/tdata/**/*.*csv

output.elasticsearch:
  hosts: ["https://localhost:9200"]
  username: "admin"
  password: "admin"
  ssl.verification_mode: "none"
```

To process the logs and send them to the dashboard, I ran the following Filebeat command:

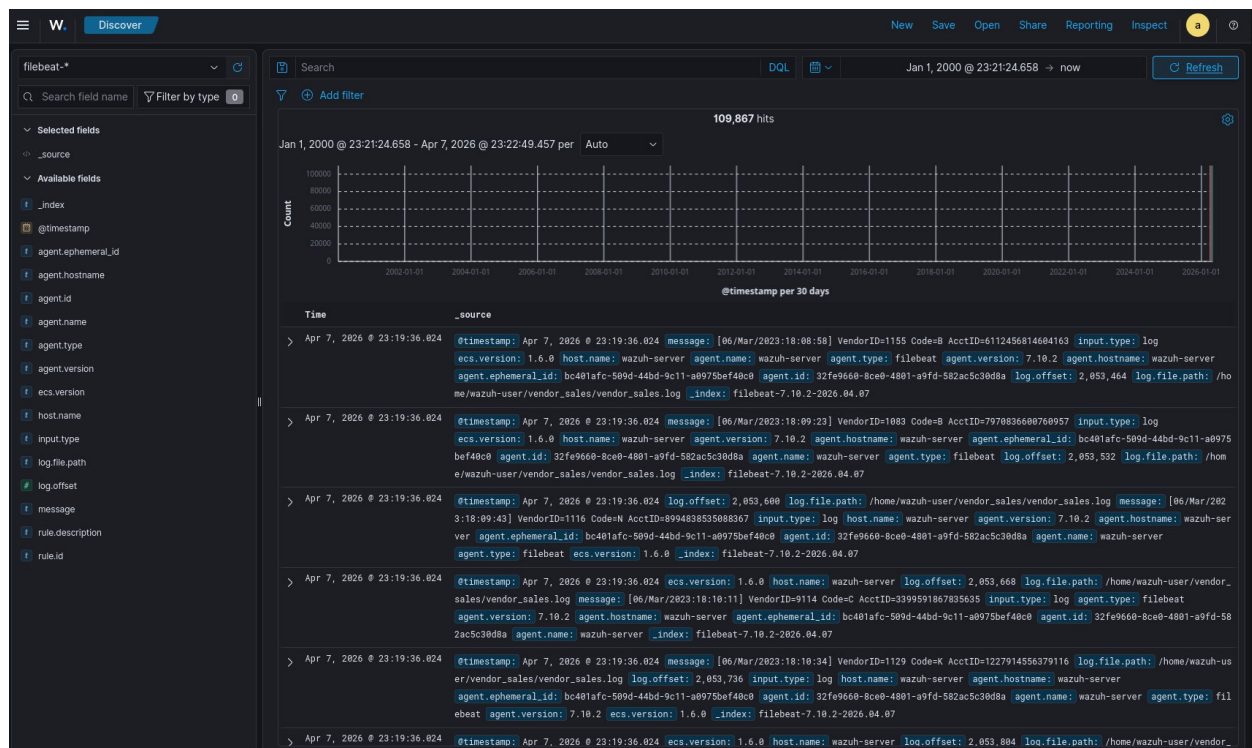
```
/usr/share/filebeat/bin/filebeat -c ingest.yml -e
```

Verifying the Data

Once the command finished,

I returned to the Wazuh dashboard in my browser and navigated to Dashboard Management > index patterns, and created a new filebeat-* index for the new logs to be displayable.

Afterwards, I returned to the Wazuh dashboard in my browser and navigated to Explore > Discover. I set the time range to Absolute with a start date of January 1, 2000, to ensure all historical data was captured. I typed * into the search bar and pressed Enter, confirming that the logs were successfully loaded and ready for querying.



Summary:

I successfully deployed and configured a local Wazuh SIEM environment using Vmware. I established an ssh connection and used wget and python server to securely transfer local files into the VM, created an ingestion configuration file, and used Filebeat to push the log data into the SIEM. Finally, I verified the data in the Wazuh dashboard, giving me a fully functional, locally controlled SIEM ready for security querying and threat investigation.